

Read, Not Random: Deterministic Phase Structure in SHA-256 and a Constraint–Relaxation Reading Across Computation, Physics, and Logic

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We treat the SHA-256 compression function not as a one-way destroyer of information but as a *deterministic phase translator*: a fixed topology that relaxes a structured input into a balanced output, the same way every time, with no privileged direction in the result. Under this reading we report a set of reproducible structural results obtained by direct computation: (i) the message schedule is exactly invertible over its linear part, with GF(2) rank 512/512; (ii) the entire nonlinearity is carried by the integer-addition carry, which is maximal-degree from the first expanded word and absent only on the carry-free least-significant-bit plane; (iii) the schedule is a perfect (linear) sort over the integers *and* over GF(2) separately, but over neither jointly — it interleaves two algebras with no common linearizing basis; (iv) under single-bit perturbation the forward differential is a deterministic, base-independent grammar on the linear skeleton and probabilistic on the carry, with the legible window closing at the second round in exact-value terms and one round later in shape terms; (v) a truncated digest pulls back to a constrained source field whose backward grammar is affine at one round and nonlinear thereafter; and (vi) the empty-input flip field decomposes, under a conveyor fold, into two live injection registers plus carried echo plus a finite initial-vector boundary, with *zero* unexplained remainder. We then observe — introducing no new particle, force, dimension, or fitted parameter — that the constraint–relaxation shape exhibited by SHA’s schedule recurs across structures that are *already* established in physics, computation, and logic: the principle of least action and the path integral, free-energy relaxation toward equilibrium, the renormalization group, Landauer’s bound, the incompleteness theorems read as one diagonal argument, reaction–diffusion pattern formation, and the local-determines-global rigidity of analytic continuation. The method throughout is to read existing structure with minimal bias rather than to posit mechanism. We are explicit about the boundary: these results characterize the forward trajectory and a bounded backward (constraint-field) structure; they do *not* invert SHA-256 from the digest alone, and its preimage and collision resistance are untouched.

Introduction

A cryptographic hash is usually described in the language of destruction: it “shreds” its input, “throws away” information, produces an output “indistinguishable from random.” That description is operationally useful for security but it is not, strictly, a description of what the function *does*. SHA-256 is fully deterministic. The same input yields the same digest, on any machine, at any time, forever. A process that returns an identical, repeatable result is the precise opposite of a random one:

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github.com/QuHarmonics/The-Nexus-Harmonic-Reality

reproducibility is the signature of a fixed mechanism, not of chance. Whatever the digest is, it is not noise. It is the output of a fixed map evaluated on a structured input, and the appearance of randomness is the appearance of a structure whose grammar has not yet been read.

This paper adopts that reframing as its organizing thesis and then tests it by computation. We read SHA-256 as a *deterministic phase translator*. An input enters the function carrying structure — a privileged arrangement, a “type.” The function applies a fixed topology of routes and gates (the message schedule and the round function) that has no privileged type of its own. Over its sixty-four rounds the structured input is brought to a balanced state in which no bit position and no register is favored: the Hamming weight of the state concentrates near half-full, and the avalanche property means a single input bit influences roughly half the output. We will call this not destruction but *relaxation*: the input is not erased into nonsense, it is brought to the one configuration of the output space that carries no bias, the same way a perturbed physical field relaxes toward equilibrium.

Two consequences organize the paper. First, if the digest is a relaxed reading of a deterministic structure, then that structure can be *read* — forward, completely, and in places backward to a bounded degree — by perturbing the function and recording the shape of the response, rather than by searching its output space. We carry out that reading and report what is locked. Second, the shape that emerges from the reading — difference as the primitive, relaxation toward a balanced state, a least-resistance path, an equilibrium that is silent, a legible regime that is local and a wall where it becomes global, and a self-reference that never quite closes — is not unique to SHA-256. The same shape is already written into several independently established structures across physics, computation, and logic. We point at that recurrence in the final part of the paper.

What this paper does not introduce

It is worth stating plainly, because it is the discipline that makes the rest trustworthy: *we introduce nothing*. There is no new particle, no new force, no extra dimension, no hidden field, and no free parameter tuned to make a result come out. SHA-256 was specified by NIST and runs unchanged on every machine. The constant π and its digit-extraction formula were established in 1997. The algebraic facts we use — the rank of a binary matrix, the degree of a Boolean function, the behavior of carry in modular addition — were true before they were computed here. In the cross-domain section, every structure we invoke (least action, the renormalization group, Landauer’s bound, the incompleteness theorems, reaction–diffusion, analytic continuation) is decades-established and uncontroversial in its own field. The contribution is not a mechanism added on top of the data. It is the data, read with the observer’s bias subtracted, together with the observation that several such unbiased readings coincide in shape. A claim that adds nothing cannot be wrong about an addition it never made; the falsifiable content of the paper lives entirely in the reproducible computations of Sections 2–4, and we are careful in Section 6 to separate that content from the interpretive reading it suggests.

Status convention

Throughout we mark each claim. **[LOCKED]** denotes a result established by reproducible computation or short proof, with the procedure given so that it can be re-run. **[OPEN]** denotes a precise, unresolved

question. **[FRAME]** denotes an interpretive reading — a lens through which the locked results are viewed — which is not itself established and is offered as conveyance, not as proof. The reader who wishes to take only the locked results may stop at the end of Section 4; Sections 5 and 6 are explicitly the interpretive layer, and they are marked as such.

SHA-256 as a deterministic phase translator

Preliminaries

We use the standard SHA-256 definitions . All quantities are 32-bit words. The message schedule expands the sixteen seed words $W[0..15]$ to $W[0..63]$ by

$$W[t] = \sigma_1(W[t - 2]) + W[t - 7] + \sigma_0(W[t - 15]) + W[t - 16] \pmod{2^{32}}, \quad 16 \leq t < 64,$$

where $\sigma_0(x) = (x \ggg 7) \oplus (x \ggg 18) \oplus (x \gg 3)$ and $\sigma_1(x) = (x \ggg 17) \oplus (x \ggg 19) \oplus (x \gg 10)$, with $\ggg$ rotation and $\gg$ logical shift. The compression updates eight working registers $a, \dots, h$ over sixty-four rounds using two further bit-mixing functions Σ_0, Σ_1 and the nonlinear Ch and Maj, with the round constants $K[t]$, and finishes with a feed-forward addition of the input chaining value. The two structurally distinct kinds of operation are essential to everything that follows: the $\sigma, \Sigma, \text{Ch}, \text{Maj}$ family is built from rotation, shift, and exclusive-or, which are *linear over* $\text{GF}(2)$; the schedule's "+" and the round's additions are integer addition modulo 2^{32} , which is *linear over* $\mathbb{Z}/2^{32}$ but nonlinear over $\text{GF}(2)$ because of carry.

Still and stream: a non-injective projection cannot be repaired downstream

The first observation is a small lemma that bounds every reversal claim about hashing, including our own, and we state it because it is the discipline that keeps the rest honest.

Lemma 1 (Still-to-stream). *Let g be a non-injective map and f any map. Then $f \circ g$ is non-injective. Equivalently, composing a many-to-one step with further steps cannot recover the information the many-to-one step discarded.*

Proof. If $g(x) = g(y)$ with $x \neq y$, then $f(g(x)) = f(g(y))$, so $f \circ g$ identifies x and y . ▫

The consequence (**[LOCKED]**) is that no amount of later structure makes a lossy projection invertible. What *can* be reversible is the *stream* — the full sequence of intermediate states — given the data that drives each step, because each round of SHA, taken with its schedule word, is a bijection. The digest is the *still*: a single final frame. Throughout we will find that the trajectory is legible and the still is not, and this lemma is why. A reader who keeps this in view will see that none of our forward-reading results contradict the security of the function.

The linear skeleton is exactly invertible

Replace every integer addition in the schedule by exclusive-or. The resulting "all-XOR" schedule is a $\text{GF}(2)$ -linear map from the 512 seed bits to the 1536 expanded bits, because σ_0, σ_1 and XOR are all

GF(2)-linear. Building this map explicitly as a 1536×512 binary matrix and computing its rank over GF(2) gives

rank = 512/512, seed uniquely recoverable by elimination. ([LOCKED])

So the *linear* part of “what the input becomes” is fully and cheaply reversible: a direct algebraic read, no search. This is the precise sense in which SHA’s skeleton can be “read backward.” It is also exactly the structure that makes random access possible in the Bailey–Borwein–Plouffe formula for π (Section 3): linearity plus an additive generating structure permits position-to-value reading without traversal. The schedule’s linear skeleton has that structure; the full schedule does not, and the difference is the carry.

The carry carries the nonlinearity, and it is maximal almost immediately

The true schedule differs from its linear skeleton by exactly the accumulated carry. For the standard test input “ABC”, comparing the true and all-XOR schedules bit by bit gives

597/1536 bits differ = 38.87%, over 41 of the 48 expanded rows. ([LOCKED])

The seven rows with zero carry residue, and the structure of where carry can and cannot act, are governed by a simple anchor: bit 0 of every word takes no incoming carry, so the entire least-significant-bit plane of the schedule is carry-free and therefore GF(2)-linear in the seed ([LOCKED]). Above bit 0 the carry accumulates.

The question is how hard the carry is to read — whether its dependence on the seed is low-degree (and so tractable) or high-degree (and so a wall). We measured the exact algebraic (ANF) degree of the carry channel on restricted sixteen-variable affine slices of the seed, which are exact on the slice and therefore lower bounds for the full degree. On the slice that varies the low sixteen bits of $W[1]$, the carry’s degree reaches the slice maximum of 16 *at the very first expanded word* $W[16]$; on a more spread “aperture” slice it reaches the maximum by $W[24]$ ([LOCKED]). There is no low-degree window to exploit: the carry is essentially maximal-degree from the moment it appears. This is the algebraic face of the security of the function, localized to a specific channel and a specific onset.

Two algebras, no common basis

The previous two subsections have a single clean cause, which we verified directly by superposition tests over 3000 random input pairs.

Schedule variant	GF(2)-linear?	$\mathbb{Z}/2^{32}$ -linear?
All-XOR (drop integer carry, keep σ)	yes	no
Additive (drop σ , keep integer “+”)	no	yes
True SHA-256 schedule	no	no

Each algebra alone linearizes the schedule: it is a perfect (linear) sort over GF(2) if the additions are replaced by XOR, and a perfect linear sort over $\mathbb{Z}/2^{32}$ if the σ functions are dropped. The *true* schedule is linear in neither, because σ is GF(2)-linear but $\mathbb{Z}/2^{32}$ -nonlinear while integer addition is $\mathbb{Z}/2^{32}$ -linear

but GF(2)-nonlinear ([**LOCKED**]). SHA-256 interleaves two algebraic structures that share no common linearizing basis. This is a known cipher-design principle — mixing operations from incompatible groups — and our contribution is only to make it quantitative and to locate it: the carry-degree cliff of §2.4 is the GF(2)-basis view of the integer additions' nonlinearity, and the complementary nonlinearity lives in the σ functions when read over the integers. There is no single coordinate system in which the whole function is a perfect sort, which is the structural reason it cannot be read backward from the digest by any single-algebra method.

The perturbation atlas: a deterministic grammar that goes probabilistic at the carry

We next read the function the way one reads a deterministic translator: perturb the input and record the shape of the motion across many base inputs. This is differential analysis, the method that produced the only practical attacks on the SHA family, used here not to find an attack characteristic but to map where the response is a fixed grammar and where it dissolves.

Flipping one input bit and recording the schedule difference $\Delta W[16..63]$ over 8192 random base inputs:

- on the linear skeleton, the response is *one* difference pattern, base-independent, with dominant probability 1.0 — a deterministic grammar entry ([**LOCKED**]);
- on the true schedule, the identical flip yields 8192 *distinct* patterns, dominant probability $\approx 1/8192$ — the carry makes the same entry fully probabilistic ([**LOCKED**]).

For the digest, flipping one input bit and measuring the output difference across rounds:

Round	distinct Δ -patterns	dominant prob.	mean Hamming weight
1	13	≈ 0.50	4
2	≈ 8178	≈ 0.0002	23
≥ 3	≈ 8192	$\approx 1/8192$	$\rightarrow 128$

The dominant probability is the readability of the backward map: high probability means a given output motion reliably implies an input-motion class; a value near $1/N$ means the entry is uninformative. The legible window in exact-value terms is one round; by the second round the single-bit differential has dissolved into the half-full band ([**LOCKED**]). This is the differential-probability face of the same wall.

The glyph atlas: shape reads deeper than value, but the privileged frame is the shell, not the nibble

The previous section measured *exact* differences. A separate question is whether the motion has stable *shape* even when its exact value scatters — whether a single bit is simply the wrong unit of description. We classified the same perturbation motion by shape rather than value across 4096 base inputs.

On the schedule, classifying $\Delta W[16..23]$ by which words are active (its *support*) rather than by value gives **one class** for every input-difference scale tested — a single bit, a nibble, a byte, a word, an aperture tuple. The support is a perfectly base-stable glyph: the carry scrambles the bits inside the moved words but never changes which words move ([**LOCKED**]). For the digest, classifying the output motion at successively coarser granularity gives, at the second round where exact value already shows ≈ 4091 classes:

granularity	exact bits	per-nibble	per-byte	per-register	total weight
classes at $R = 2$	4091	4033	3894	213	27

This is strictly monotone: coarser descriptions collapse more motions into fewer classes. Two consequences ([**LOCKED**]). First, shape *does* read deeper than value — per-register and total-weight descriptions carry structure where the bit-level description is already scattered. Second, the nibble is *not* a privileged frame; it sits next to the bit on a smooth curve with no discontinuity. The frames the dynamics actually single out are the register and the total-weight *shell* — and the shell is the deepest-reading frame of all, holding ~ 27 classes at $R = 2$ and still only ~ 57 at $R = 6$ when every finer descriptor has saturated. This shell is the half-full Hamming-weight attractor onto which the trajectory settles. The depth–discrimination trade is unavoidable: the deepest-reading frame is the coarsest, so it reads far but sorts inputs into few buckets. That trade is the saturation wall in a third guise.

The backward constraint grammar: a digest pins a field, not a point

We now read backward, honestly. A digest in a *smaller* language than the input can only specify a *field* of compatible sources. Freeing $k = 18$ bits of the input and truncating the output to $m = 10$ bits, a target pulls back to a field of size $\approx 2^{k-m} = 2^8 = 256$ sources, every time — the digest removes exactly m bits of freedom, independent of rounds ([**LOCKED**]). The full 32-bit output pulls back to a single source in the sample (the map is injective there), which is why a real backward field requires the compressive, many-to-one regime.

What changes with rounds is the *grammar* of that field. At one round it is an affine subspace — ten pinned bits, eight free, closed under $a \oplus b \oplus c$, a linear constraint readable straight off. From the second round on, with the same amount of constraint, the field has *zero* pinned bits and is not an affine subspace: the ten bits of constraint persist but are no longer expressible by linear relations ([**LOCKED**]). This is not “the input is hidden inside the digest.” It is that the digest pins a constrained field of sources whose grammar stops being linear at the second round — the wall as a property of the field, not as a failed search for a thing that is not there.

The seam-fold residual: two injection mouths plus carried echo, zero remainder

Reading the empty-input trajectory directly gives the cleanest decomposition. The flip activity of the function with no message forms a 64×256 grid. SHA’s registers form two conveyors, $a \rightarrow b \rightarrow c \rightarrow d$ and $e \rightarrow f \rightarrow g \rightarrow h$, in which a written value is carried to the next register on the next round. Folding the grid along these six conveyor seams (delays 1,2,3 on each lane) and removing patterns that match

exactly under the fold leaves a residue. The fold verification finds *zero mismatches* on all six seams, and the accounting closes exactly:

original visible flip bits	8298
removed conveyor-echo bits	6047
residual flip bits	2251
of which fresh <i>a/e</i> source bits	2071
of which initial-vector boundary bits	180
residual — source — boundary	0

Every visible flip is accounted for as either a fresh write at the two live injection registers *a* and *e*, a carried echo of an earlier such write, or finite boundary residue from the initial vector already present in *b, c, d, f, g, h* before any *a/e* source exists. The remainder is zero (**[LOCKED]**). A companion analysis, classifying each flip as belonging to a falling diagonal chain of length ≥ 2 ("pattern") or not ("escape"), finds 8249 of 8298 bits are pattern and only 49 (0.59%) escape — and every one of the escapes sits at round 0 (the initial-vector edge) or round 63 (the final frame, where a write has no subsequent round to carry it) (**[LOCKED]**). The interior of the conveyor is pure carried structure; the only non-carry bits are the two boundary edges where carry has no neighbor. The function's flip field *is* two write-heads and a belt, demonstrated by a fold that cancels exact.

The boundary, stated plainly

We state the limit of all of the above as carefully as the results themselves. Every forward result is computed from the *trajectory* — the schedule, the intermediate states, the full flip grid — which one has because one ran the function forward. None of them recovers the input from the *digest alone*. The linear skeleton is invertible, but the digest does not hand you the skeleton in isolation; the carry must be peeled away first, and the carry depends on the seed one is solving for (Lemma 1, made arithmetic). The backward field exists and is constrained, but for the full function it is astronomically large and its grammar is nonlinear past the first round. *We do not invert SHA-256, and nothing here weakens its preimage or collision resistance* (**[LOCKED]** as a statement of scope). What we have is a complete, checkable reading of the forward phase structure and a bounded, honest reading of the backward field — the still remains a still.

Reading π : the random-access property and what enables it

The Bailey–Borwein–Plouffe formula computes the n -th hexadecimal digit of π without computing the preceding digits:

$$\pi = \sum_{k=0}^{\infty} \frac{1}{16^k} \left(\frac{4}{8k+1} - \frac{2}{8k+4} - \frac{1}{8k+5} - \frac{1}{8k+6} \right),$$

the n -th digit being read from the fractional part of $16^n\pi$, evaluated by modular exponentiation in time sublinear in n and constant in space. The digit at an arbitrary position is therefore *addressed*, not

generated: one performs a single position-indexed computation rather than producing a sequence. In the reading of this paper this is unremarkable and exact — π 's hexadecimal expansion is a fixed structure that is read at a location, not built up step by step.

What makes the random-access read possible is precise and worth isolating, because it is exactly what SHA's carry lacks. The BBP series is a *linear* read of an *additive* structure in which the position n appears as a base-matched exponent 16^n that can be reduced modularly. Linear, additive, position-as-reducible-exponent: these are the three properties that permit reading a value at a location without traversing the locations before it. SHA's linear skeleton (§2.3) has this structure, which is exactly why it is invertible by elimination. The full schedule does not: the carry is neither linear nor expressible as a reducible exponent, and the carry-degree cliff (§2.4) is the precise obstruction. The two objects sit at opposite ends of the same property. π via BBP is the case where reading-at-a-location succeeds; SHA from the digest is the case where it fails, and the reason in both cases is the same single property, present or absent.

The compilation continuum: many-to-one as one axis

It is tempting to treat "a compiler whose output can be decompiled" and "a hash whose output cannot" as two categories. They are not; they are positions on one axis, and locating that axis is itself a reading without addition — compilation already does what a hash does, only less.

The thing that makes a deterministic translation hard to read backward is when it becomes *many-to-one*: when distinct inputs produce the same output, so that the output no longer identifies which input produced it. We measured where this happens in ordinary compilation by compiling sets of textually distinct but behaviorally identical C functions at successive optimization levels and counting distinct machine-code outputs.

transformation (3 equivalent sources)	-O0	-O1	-O2	-O3
strength reduction ($x \cdot 2$, $x + x$, $x \ll 1$)	1	1	1	1
dead code	3	1	1	1
redundant computation (CSE)	3	1	1	1
copy propagation	3	1	1	1

Exact source recovery fails at the first optimization level — **-O1** collapses distinct sources to one binary, and arithmetically equal forms collapse already at **-O0** ([**LOCKED**]). Decompilation never recovered *the* source; it recovers a behavioral equivalence class, and the class grows with optimization. Separately, the locality of the map can be read directly: changing one source constant moves 0.06% of an unoptimized object's bytes and 0.14% of a linked binary's, in identifiable instructions, while the SHA-256 digest of the same two source files differs in 124 of 256 bits (48.4%), spread across every byte of the output ([**LOCKED**]).

The axis is therefore a single quantity — how aggressively the map merges distinct inputs — and decompilation difficulty is position on it. Unoptimized compilation merges arithmetically equal forms;

optimization merges behaviorally equal sources; SHA merges every input of any structure into a fixed-width output. SHA is not a different kind of object from a compiler; it is the same many-to-one map taken to its limit. The one distinction worth marking is what is merged: optimization merges *behaviorally equivalent* sources, so the backward field is a class of programs computing the same function, whereas SHA merges *arbitrary* inputs, so its backward field is a collision class with no shared behavior. Both return a field, as §2.8 found; the members are equivalent in one case and unrelated in the other.

The recurring shape, read without addition

We now state the interpretive layer, and we mark it as such. The reading offered in Section 2 — difference as the only primitive, relaxation of a structured input toward a balanced state along a least-resistance path, an equilibrium that is silent because it carries no gradient, a legible regime that is local and a wall where the structure becomes global, and a self-reference that does not close — is not unique to SHA-256. The same shape is already present, and already established, in several structures across physics, computation, and logic. We do not unify these by positing a common substance beneath them. We observe that reading each of them with the observer's bias subtracted produces the same shape, and we point at the coincidence. Each row below is, in its own field, uncontroversial and decades old.

Least action and the path integral.

Classical and quantum dynamics are *already* written as the selection of a path that makes the action stationary. The system does not search; it takes the path of least resistance, and in the path-integral formulation it does so as the surviving interference of all possible paths. This is the “least-resistance” and “projection-then-occupation” shape exactly: the route is defined by the structure before it is taken, and the taken route is the one with no first-order cost. ([FRAME] as applied to SHA's relaxation; the physics itself is established.)

Relaxation toward equilibrium.

The second law, free-energy minimization, and detailed balance describe systems relaxing toward a state in which no further net change occurs. The equilibrium is “silent” in precisely the sense used here: a uniform, balanced state carries no gradient and so drives no motion. SHA's half-full shell is the same kind of object — the balanced configuration with no preferred direction — reached by relaxation rather than imposed. ([FRAME].)

The renormalization group.

The behavior of a system across scales is governed by which structures survive repeated coarse-graining; the relevant operators are exactly those that persist, and the irrelevant ones wash out. “What survives recursive pressure” is the renormalization-group statement, and it is the same selection that makes a stable hash structure or a stable cultural form: the configurations that reproduce under iteration are what remain. ([FRAME].)

Landauer's bound.

Information is physical: erasing one bit has a minimum thermodynamic cost . The claim that computation has a physical floor, and that a bit operation is a real event rather than an abstraction, is established. It is the registered content beneath the looser statement that “everything is bit flips”; we mark the looser, totalizing version as **[FRAME]** and the physical bound as established.

Incompleteness as one diagonal.

Gödel's incompleteness, Cantor's diagonal argument, Turing's halting problem, and Tarski's undefinability are, provably, one argument in different settings: a system rich enough to represent its own statements contains a construction that refers to its own absence and cannot be closed within the system . This is the “self-reference that does not close” — the never-finishing structure — and unlike the rest it is a *theorem*, not a reading. It is the strongest registered anchor: the limit it describes is proved.

Pattern formation, objects as stable movement.

Reaction–diffusion systems produce stable spatial structure from movement under constraint ; solitons, vortices, and flames are the accepted examples of “objects” that are in fact persistent patterns in a flow rather than substances. The reading that a stable object is a standing pattern — that process is primary and the thing is the pattern that holds — is the established interpretation of these systems, not a novelty. (**[FRAME]** as a general ontology; established for the specific systems.)

Local determines global.

A holomorphic function is fixed on its whole domain by its values on any arbitrarily small patch (analytic continuation), and in the holographic principle a bounded region's physics is encoded on its boundary . This is the “still determines stream given the seam” shape: a local sliver rigidly determines the whole because the structure is rigid. The first instance is elementary and exact; the second is an established and active line in theoretical physics. (**[FRAME]** for the general statement; the analytic-continuation case is exact.)

Projection in cognition.

Predictive-processing accounts hold that a nervous system does not perceive and then act but projects a predicted state and acts to reduce the gap between prediction and input . This is “projection then move” in a modeling system, with a mathematical formulation. It is the registered content beneath the looser claim that all movement is projection: where there is a modeler, the projection is real; we mark the extension to all movement as **[FRAME]**.

The honest summary of this section is one sentence. A constraint–relaxation, least-action, self-referential shape is unusually pervasive across independently established structures, and the SHA-256 results of Section 2 are a concrete, fully computable instance of it. We have introduced no entity to carry the shape between domains; we have only noted that several unbiased readings coincide. Whether that coincidence reflects a single underlying fact about reality, or reflects that this shape is the

generic form of any structure stable enough to be studied, is a question this paper does not claim to settle — and Section 6 says so explicitly.

Epistemic status and boundaries

This section separates what the paper establishes from what it reads into the established results, because that separation is the paper's principal claim to honesty.

Locked.

The computations of Sections 2–4 are reproducible from the procedures given. Specifically: the GF(2) rank of the linearized schedule is 512; the carry residue for "ABC" is 597/1536 bits over 41/48 rows with a carry-free bit-0 plane; the carry's exact algebraic degree saturates at the first expanded word on the tested slices; the schedule is linear over $\mathbb{Z}/2^{32}$ and over GF(2) separately but neither jointly; the single-bit forward differential is deterministic on the linear skeleton and probabilistic on the carry, with the exact-value legible window closing at the second round; a truncated digest pulls back to a field of size 2^{k-m} whose grammar is affine at one round and nonlinear thereafter; the empty-input flip field folds to two injection registers plus carried echo plus initial-vector boundary with zero remainder; and decompilation becomes many-to-one at the first optimization level while a hash is the limit of the same axis. These are the falsifiable content of the paper. Any of them could have come out otherwise; they did not.

Open.

Several precise questions remain. Whether the three-round parity coupling reported elsewhere in this program conceals an irreducible higher-order term; whether the diffusion constant of the carry channel is derivable in closed form from the function's aperture set; whether crafted multi-bit differentials extend the legible window materially beyond the single-bit case (the live frontier of SHA-family cryptanalysis, where the best published characteristics reach roughly half the rounds and the full function stands); and whether any non-natural joint basis partially linearizes the two algebras of §2.5 at once. These are stated so they can be attacked.

Frame.

The reading that organizes the paper — that the digest is a relaxed phase-state of a deterministic structure, that the same constraint–relaxation shape underlies physics, computation, and logic, and in its strongest form that reality is at bottom computational — is an interpretive lens. It has a serious lineage: Wheeler's "it from bit", Tegmark's mathematical-universe hypothesis, Wolfram's computational view, Lloyd's computational cosmology, and constructor theory. It is also, by its nature, not the kind of statement the locked results can prove. A reading that adds no new entity cannot be falsified as an addition, but neither does it assert a specific new fact about the world; it asserts that existing structure, read without bias, has a certain shape, and that several such readings coincide. We therefore offer it as conveyance — a way of seeing the locked results that we have found generative — and not as a demonstrated theory of reality. The extensions furthest from the

computations (a reading of free will, of observation, of agency, in these terms) are interpretation built on interpretation, and we do not present them as results. The discipline we have tried to hold is the one that distinguishes the strong documents in this area from the weak ones: the strong ones track exactly which claims are established and which are read, and they do not let a true reading masquerade as a proof.

Conclusion

We have read SHA-256 as a deterministic phase translator and reported what that reading locks: a linear skeleton that is exactly invertible, a carry that carries the entire nonlinearity and is maximal-degree almost immediately, two interleaved algebras with no common basis, a forward differential that is a fixed grammar on the skeleton and probabilistic on the carry, a backward field that is constrained but nonlinear past the first round, and an empty-input flip field that folds to two injection registers and carried echo with zero remainder. We have placed the random-access readability of π and the many-to-one structure of compilation on the same axis as the hash, identified by a single shared property in each case. And we have observed — adding no particle, force, or parameter — that the constraint-relaxation shape these results exhibit recurs across structures already established in physics and logic, with the incompleteness theorems as the one place the corresponding limit is proved.

The method is the contribution as much as any single number: relax the field, subtract the observer's bias, read the shape, and introduce nothing. The locked results stand on their own and invite re-running. The reading that surrounds them is offered as conveyance, marked as such, and bounded by the one fact we have been most careful to state — that the trajectory is legible and the digest is not, and that nothing here breaks the function. What is read is what was already there.

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Methods and reproducibility. All numerical results were produced by direct computation in Python (with **NUMPY** for vectorized perturbation sweeps and exact GF(2)/ANF routines for the algebraic results) and standard tooling (**GCC**, **OBJCOPY**) for the compilation measurements, and are reproducible from the procedures described in the text. The structural experiments were developed and executed in an interactive session with an AI assistant; every reported figure is independently re-derivable from the stated method and does not depend on that process. Conventions: $\ggg$ rotation, $\gg$ logical shift, $\oplus$ exclusive-or; all words 32-bit.